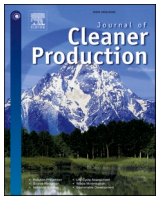
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Walking the tightrope: Can integrated decision support transform business sustainability?

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ABSTRACT

Achieving the Sustainable Development Goals (SDG) Agenda 2030 requires accelerated reorientation and investment towards social and environmental goals, with sophisticated methods to resolve inherent trade-offs across goals and improvement on how companies define, measure and pursue social and environmental value. We use action research to test and evaluate the ability of an integrated assessment tool to support balanced assessments and decision-making across economic and environmental goals in two large corporations. Following characterisation of Environmental, Social and Governance practices, Multi Criteria Decision Analysis (MCDA) was applied to three use cases and the process evaluated and compared to business-as-usual decision-making processes. MCDA was intuitive, widely understood, models were easily developed by working with relevant staff to identify alternative options for ranking, appropriate criteria, measures, and parameters. MCDA was effectively applied to a wide range of decision problems related to materiality ranking, investment, and capability prioritisation. The process drove broader stakeholder engagement, revealed and challenged implicit assumptions and preferences on cost minimisation, improved robustness and transparency of decision-making, and, through integration of non-economic criteria, shifted favoured alternatives in all use cases. We pinpoint challenges and opportunities for broad-scale adoption of MCDA.

1. Introduction

Greenhouse gas emissions are set to rise 14% over the current decade as a result of persistent upwards trends in global GDP per capita and population despite improvements in economic energy intensity and energy CO_2 intensity (Masson-Delmotte et al., 2021; United Nations, 2022). With similar trends for biodiversity (Dasgupta, 2021) and global material use (Hickel and Kallis, 2020; Schandl et al., 2016), it seems clear that achieving the environmental SDGs alongside economic growth as ambitioned in SDG 8 requires explicit trade-offs and consideration of relative and absolute degrowth concepts. In response to this dilemma, several counterpoints to mainstream economic theory have re-emerged, questioning the ability of current governance approaches to successfully decouple natural resource consumption and environmental impacts from economic growth (Parrique et al., 2019), questioning the 'sustainable development' paradigm, and calling for altogether different approaches to accelerated and transformational change (Coscieme et al., 2019). A road untravelled, the ideas put forward for this paradigm change include a swathe of policy measures to ensure the representation of social and environmental values in the marketplace, from smarter tax regimes to the regulation of advertising to dampen consumerism of goods; accounting for the externalities of GHGs, the direct and indirect co-benefits of climate action, far more sophisticated assessment of risk, steps to reduce welfare dependence on economic growth (Corlet Walker et al., 2021); and substantial targeted investment in innovation of technology and services that will allow the reorganisation of manufacturing into closed-loop production systems that conserve natural capital (Stern et al., 2022).

Against this background agenda for accelerated and transformational change, key challenges to achieving Agenda 2030 remain around resolving inherent trade-offs across goals, balancing conflicts of interest, aligning on targets, and ensuring adequate investment - challenges that will be amplified as climate change claims increasing investment in disaster response. At community and organisational levels, these challenges manifest in the form of 'tensions' or competing environmental and economic objectives where explicit choices must be made in pursuit of green growth (Ranängen et al., 2018; Van Der Byl and Slawinski, 2015). For example, should profit be retained as earnings to

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invest into new products and services, increased market share, maintenance and expansion of capital assets, shareholder dividends, employee wages and incentives, societal good, or emissions reductions? These are real dilemmas that executives, boards, and shareholders face when applying Environmental Social and Governance (ESG) principles in practice. However, by and large, shareholder returns continue to form the bedrock of decision-making for capital allocation and enterprise value creation (Câmara and Morais, 2022). Both in literature and practice, economic indicators are still regarded as the benchmark for corporate performance, with investments in environmental and societal sustainability often seen as necessary costs, or as means to improve on corporate economic performance, rather than sustainability being integral to corporate performance itself.

This study examines the tension between SDGs 8 (Decent Work and Economic Growth) and 13 (Climate Action) as it manifests in two large New Zealand corporate organisations and explores the extent to which corporate decision-making tools and frameworks inadvertently pit prosperity against the planet. Through observation and action research, we explore the challenges of prioritising and implementing sustainability efforts and achieving progress on conflicting targets, applying integrated decision-support methods in situ to aid corporate sustainability practices. In what follows we outline the purported role of integrated decision support in transformative business sustainability, reviewing what is known about the preconditions for its effectiveness, challenges or limitations, and identify gaps in evidence. Building on the evidence gap, we set out our research questions and methodology, our results, analysis and conclude with pragmatic recommendations.

2. Literature review

Criticisms of corporate sustainability efforts are wide ranging; from discrepancy between professed and actual commitment to corporate sustainability and rebranding of business-as-usual process efficiency gains ('green-washing') (Lee and Raschke, 2023), self-selection into voluntary reporting systems (Doda et al., 2016), lack of enforcement of existing standards, to substantial ambiguity of actual societal outcomes (Johnston et al., 2007), where current action amounts to "merely reducing unsustainability" and existing reporting mechanisms obstruct real action (Flammer, 2013; Hoffman and Ehrenfeld, 2013). In-depth consideration and integration into business strategy, operational management, organisational structure, product and process design is often lacking (Hoffman and Ehrenfeld, 2013; Milne and Gray, 2013), with little empirical evidence of 'deep' culture change (Harris and Crane, 2002). Capital investments to grow enterprise value are nearly universally benchmarked by forecasting and discounting cash flows using interest rates that represent debt and equity capital costs alone, failing to account for long term environmental and social costs. Câmara and Morais (2022) point to shareholder primacy as a key barrier for the adoption of more holistically responsible practices and argue that gaps exist in knowledge, accountability, leadership, execution, and disclosure at the political, regulatory, legal, and corporate governance level. Taken together, it is not surprising that less than 1% of S&P 500 firms have successfully decoupled net revenues from natural capital costs (Greenbiz, 2019). Despite the varying degrees of alignment with opposing paradigms of weak to strong sustainability across the broader literature on business and sustainability (Belmonte-Ureña et al., 2021) - the emerging consensus is that more must be done by the private sector to redefine the status quo and reform organisational decision making towards more holistic long-term outcomes.

For business, transformative sustainability necessarily involves action taken outside traditional and hierarchical organisational structures through multi-stakeholder collaboration (Jacobsen et al., 2020; Kuenkel et al., 2020; Ritz and Rimanoczy, 2021). As such, corporate leaders in sustainability proactively engage with multi-stakeholder groups, are transparent about outcomes, incorporate non-economic benefits into strategic planning processes, have integrated performance incentives underpinned by a strong culture, and prioritise issues based on materiality rather than cost-benefit and potential returns alone (Pandey et al., 2019). Furthermore, business activity can become sustainable when accompanied by activism, collaborative value creation, democratic governance, end-to-end product lifecycle assessment and responsible consumption (Khmara and Kronenberg, 2018; Klettner et al., 2014). Implementing this requires a genuine commitment to transform, essentially deviating from short term profit maximisation as the primary and only goal (Nesterova, 2021). However, it remains unclear how business should go about resolving tensions between responsible production and consumption vis-à-vis maximisation of shareholder value, or what the sustainability transition implies for business performance and returns (Nesterova, 2020).

To help us further conceptualise how organisations manage 'tensions' across competing environmental and economic objectives, Van der Byl and Slawinski (2015) distinguish different approaches: win-wins (avoiding tensions by restricting practices to measures that generate economic and environmental gains), trade-offs (compromising on either economic or environmental gains where they conflict), and integrated thinking (managing tensions by shifting focus from economic to environmental gains, to give a more equal weighting to sustainability elements). While much of corporate sustainability literature reinforces a focus on win-wins, and views sustainability performance in terms of short- and long-term financial performance (e.g., see Epstein et al., 2017), integrated thinking refers to the consideration of value creation over short, medium and long-term horizons, across all the capitals that an organisation employs.

Decision support frameworks can play a key role in making non-economic impacts (both positive and negative), interlinkages and tensions more visible at organisational and supply chain levels. There are a wide range of tools available, each fulfilling different needs, from tools to help assess sustainability (e.g. tracking and visualising environmental or circular economy performance overall, to carbon footprint analysis, product life cycle or supply chain sustainability assessments, energy optimization and building efficiency) (Ness et al., 2007; Martinetti and Havas, 2021), tools to model appropriate (science-based, planetary boundary) targets (Erlandsson et al., 2023; Walenta, 2020), to a wide range of integrated assessment tools that help to evaluate and compare competing operational and management, investment or strategy decisions (Bui, 2000; Munda, 2005; Ness et al., 2007; Zavadskas et al., 2016). In the latter category, a well-established approach is Multi-Criteria Decision Analysis (MCDA), which ranks alternative options by comparing their performance across multiple incommensurable dimensions to support the evaluation of alternative decision options in an integrated way (Belton and Stewart, 2002; Greco et al., 2016). By structuring decision making along 1) *the problematic*, or the question that needs to be answered via the analysis*, 2) alternative solutions* that must be evaluated for a given problem; and 2) *criteria* that are used to assess each alternative, MCDA facilitates and promotes structured conversations between stakeholders and experts (Angelo, 2021; Cohen et al., 2019). The collaborative process of structuring the problem, alternative solutions and standards for judgements thought to be important, along with co-impacts, synergies and trade-offs, can drive clarity and transparency, and has been used to manage conflicts arising from different goals and actors (Cohen et al., 2019). One of the benefits of MCDA approaches versus traditional Cost Benefit Analysis (CBA) or risk-opportunity analysis is the integration of both quantifiable and qualitative criteria into the decision problem, allowing for inclusion of co-benefits and other difficult-to-quantify criteria that might otherwise be excluded (Ürge-Vorsatz et al., 2014).

MCDA has seen academic applications in engineering, finance, and investment portfolio management, and has a strong academic basis in energy and transport industries (Mardani et al., 2015). Although there are exceptions (e.g., see Dias et al., 2018; Ferretti et al., 2019), the literature is dominated by narrow and relatively technical applications. In practice, the use of MCDA models and other integrated assessment

tools to replace standard commercial business cases with a more holistic framework is far from mainstream (Mercure et al., 2021; Ness et al., 2007; Xu, 2020). Case studies applying MCDA in corporate setting show promise (Aktaş and Demirel, 2021; Bezerra et al., 2021; Garcia et al., 2016; Thabrew et al., 2018), but research on the practical limitations and challenges of applying MCDA in corporate decision support contexts is scant.

The broader literature on decision support suggests that if MCDA has a potential role to support better decision making on sustainability, it must be intuitive, useable, easily understood, models easily developed, with model alternatives, criteria, measures and parameters easily defined (Wong-Parodi et al., 2020), supported by relevant available data, analytical capacity and willingness of relevant stakeholders to engage in the process (Bui, 2000). It must facilitate transparent assumptions and ability to customise and apply models for different decision contexts and time horizons (Bui, 2000; Frini and Benamor, 2018; Witt and Klumpp, 2021; Yu and Du, 2022).

In summary, literature suggests that corporate sustainability efforts struggle with the prioritisation and balancing of dependencies, uncertainties and tensions between competing goals. At the same time, there is a gap in the evidence of the viability of integrated assessment tools such as MCDA in aiding development of enterprise planning and ESG efforts in holistic corporate sustainability prioritisation and implementation beyond technical or engineering choices, fuelled in part by the lack of research on the interface between decision support, organisational learning and sustainability performance. Although MCDA is widely purported to support better decision making on sustainability, we do not know for what problems, contexts, and scale it might be effective to support corporate decision making, and what are the necessary requirements for it to be relevant, useable and effective.

3. Aims and research questions

This study uses action research methods to examine the tension between economic and environmental goals in decision-making and evaluates the use of MCDA to support balanced decision-making in two large corporations, exploring the relationship between integrated assessment tools and environmental economic trade-offs in a broader context of transformative business sustainability for the first time. The overarching aim was to investigate opportunities for improvement in integrated decision-support practices in corporate settings to support the development of sustainability strategy, planning, and project level execution, focussing on the applicability of MCDA to integrate SDGs 8 and 13. Specifically we ask.

1. Can MCDA be deployed effectively as a decision-making framework for corporate sustainability efforts to aid balanced assessments and rational decision-making on behalf of stakeholders for short and long-term goals?
2. Does application of MCDA support integrated thinking across environmental and social dimensions to progress SDGs 8 and 13 simultaneously?

4. Methodology

To answer these questions, we adopt action research methodology (McNiff and Whitehead, 2009), taking the view of practitioners in corporate sustainability working collaboratively with key stakeholders and collaborators to identify relevant ESG decision challenges to be tackled, structuring the problem definitions, and building MCDA models in an iterative and interactive manner. In what follows, we explain how we tested the application of MCDA decision support models via 'use cases' at two organisations and evaluated their ability to progress more balanced outcomes towards decision making by comparing them to traditional decision-making methods (Fig. 1). We then outline selection criteria, MCDA method used, model development and software, and the limitations of this research.

Action research is a critical participatory method enabling practitioners to improve learning and practice to meet the needs of changing times and circumstances (Kemmis et al., 2014), distinct from traditional research methods in that it seeks to drive continuous improvement or change through learning, employing and testing practical solutions to problems in an iterative manner (McNiff and Whitehead, 2009). Learning is shaped by the actions and interactions of the researcher in the organisational context, and assessed in large part through critical self-reflection, but can be augmented with feedback from participants in the study.

Two organisations were selected for this study on the basis of existing relationships, trust, and observations that both organisations had well-developed sustainability strategies and capabilities, yet found the trade-offs in decision-making challenging. Both organisations have high public profiles as essential service companies in the energy and transport logistics sectors in New Zealand. Their identities have been kept confidential for data privacy and commercial market sensitivity. The practitioner researcher assumed a high-level leadership and advisory role in both organisations and was in a unique position to observe challenges in governance, execution and achievement of sustainability outcomes. She was able to observe and document current decision processes as well as participate, contribute and use the learnings in real-time to inform and improve practices. In seeking answers to the research questions, the research involved the following steps.

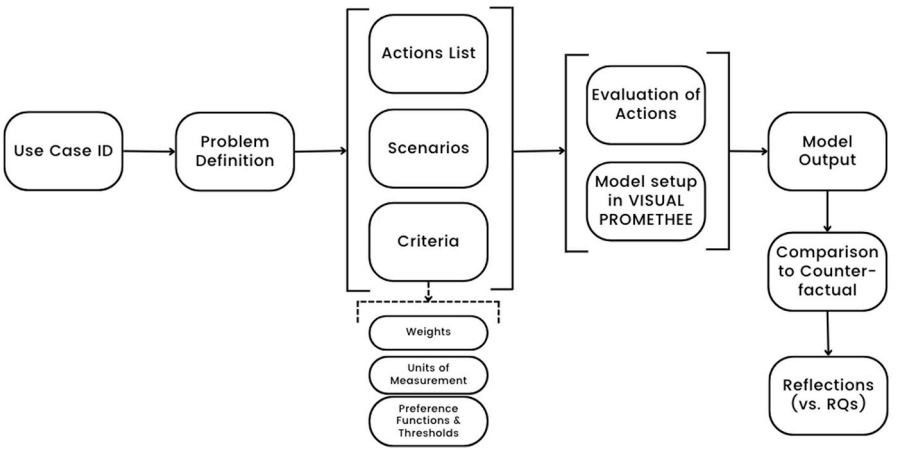


Fig. 1. Overview of the methodological approach.

1. Identification of current shortcomings and challenges in ESG practices;
2. Identification of real-world relevant use cases for decisions with implications for sustainability that could benefit from integrated assessment methods;
3. Iterative implementation of MCDA prototypes for selected use cases;
4. Comparison of counterfactual decisions generated by business-as-usual decision support methods and MCDA model outcomes;
5. Qualitative feedback from participants on the model development process, implications, benefits and shortcomings of the use cases;
6. Evaluation of MCDA development and decision support processes, drawing on data collected and applying relevant criteria.

As such, this study involved a range of different data collection methods.

- **Participatory observations in sustainability and strategy discussions** were used to identify challenges and limitations of balancing progress against economic and environmental goals, current methods of prioritisation, and opportunities for improvement.
- **Interviews and workshops** with sustainability managers were used to identify use cases to test via MCDA and build the prototypical models.
- **Counterfactual decision frameworks and outcomes** consisting of existing or documented decisions made using traditional cost-benefit analysis or other techniques that explain their basis.
- **MCDA Models for each use case** consisting of alternatives, criteria, weights, preferences, and ranking outcomes which were analysed and compared to the counterfactual decision frameworks and outcomes.
- **Interviews with stakeholders** (executives, high level decision makers and individuals outside of the organisation impacted by the decision) **and participants** (staff involved in developing the MCDA models and providing the data) to obtain qualitative feedback on model development, implications of model rankings relative to counterfactual decisions. Six individuals were interviewed; the collaborators for each use case, the sustainability lead, and the executive sponsor for the research across both organisations.
- **An action research journal** was used to capture continuous learnings through the course of the study. The journal recorded observations, rationale and learnings, feedback from collaborators, and significance of the learning. This enabled iteration, self-reflection, and evaluation to happen at every step of the process and inform subsequent actions for improvement.

Table 1 outlines the questions for self-reflection and the interview questions used to evaluate the use cases. The participants in the study were collaborators across sustainability, finance, and strategy teams whose knowledge and expertise was necessary for producing analysis as it relates to decision-making at an enterprise governance level. They played a pivotal role in the joint identification of the decision problem to be solved, construction of the alternatives, choosing the right criteria and weights, testing the application of MCDA relative to traditional decision-making processes, and reflecting on collective feedback to make recommendations for future consideration. There were twelve participants in total across both organisations.

4.1. Criteria for use case selection

To determine suitable problems for development of use cases, the model framework was explained to participants and problems jointly assessed for suitability, from the perspective of benefit to the organisation. There was a need to ensure that the selected use cases were complex enough to warrant MCDA but at the same time, simple enough to evaluate and complete within a reasonable timeframe. Use cases had to meet three criteria. First, use cases needed to contain alternative actions (choices) that were necessarily ambiguous in their values for competing economic and environmental criteria. Second, the choice of problems to be solved via MCDA needed to be able to be evaluated within reasonable timeframes, as developing the model and conducting due diligence to enable confidence in the recommendations required time and expertise. Finally, in order to determine whether MCDA improves the status quo for ESG practices, we needed a benchmark counterfactual decision to exist, or a parallel process to make such a decision needed to be active.

4.2. MCDA method, Model development and software

MCDA is a family of distinct approaches to modelling information and stakeholder preferences, compare initiatives and support decision making (Belton and Stewart, 2002). In this study we used an outranking approach as opposed to a value aggregation approach in order to allow incorporation of qualitative preferences and avoid compensation (offsetting) of one criterion against another, reducing the possibility that poor performance in one dimension is compensated by very good performance in another (Ishizaka and Nemery, 2013). A number of

Table 1
Questions used to evaluate Use Cases.

Research Questions	Effectiveness of MCDA as a DS tool for sustainability	Scaling MCDA from project to enterprise level	Contribution of MCDA towards balancing SDGs 8 and 13
Interview Questions for participants and stakeholders	• Was MCDA easy to understand? • Has it been a useful exercise, if so, how has it been useful?/Why not? • How easy was it to build the models? • How easy was it to evaluate alternatives in the models? • What would improve the process?	• Would you use this method to apply across other ESG decisions? • Does MCDA link easily into existing ESG processes i.e. materiality analysis, GRI, Integrated Reporting?	• Did MCDA improve consistency, stakeholder influence, transparency and balance of criteria versus counterfactual decisions? • Did you find the recommendations valuable and relevant?
Questions for Self-Reflection	• Was it easy to explain MCDA to the team? • Was problem structuring in MCDA suited to team needs? • Was the necessary data and human resource capacity available? • How useful was the ranking of alternatives to support the decision process?	• What are the limitations of MCDA at project and organisational level? • Can MCDA templates be used across a variety of applications? What could stay the same and what would need to change? • How often do models need to be updated?	• Do MCDA results diverge from counterfactual? • How did MCDA shape subsequent discussions and decisions? Did it reverse or change judgements or decisions?

outranking methods are available, ranging from AHP (the use of a hierarchy of from objectives to alternatives to conduct pair-wise comparisons), ELECTRE (outranking based on multi-attribute utility theory), to PROMETHEE (Preference Ranking Organisation Method for Enrichment and Evaluations in which alternatives are ranked based on pair-wise preference flows) (Opricovic and Tzeng, 2007). We used PROMETHEE on the basis that is well-established, and especially well-suited to towards problem structures where the alternatives are difficult to reconcile against each other, and scenarios where structure, transparency, and justification for recommendations are desired (Brans et al., 1986; Behzadian et al., 2010). PROMETHEE is based on a simple mathematical structure that is easy to explain to the lay person and easy to use for non-technical users, providing a visual and relational framework that can highlight conflicts, synergies and factors underlying its rankings (Bezerra et al., 2021; Munda, 2005).

Once potential use cases were identified, the exact question to be answered and/or desired rankings sought was collaboratively defined, and a schematic developed for the scenarios to be tested (*the problematic*) (Fig. 1). This was followed by an iterative identification of alternatives (*options*) to be ranked and a discussion of what level of granularity was necessary, often iteratively, until a reasonable level of specificity could be obtained that was neither too extensive (necessitating much more due diligence on evaluation) nor too high level (rendering the ranking ineffective). Alongside this process, criteria and units to evaluate alternatives and corresponding weights were constructed, followed by setting of preference functions and indifference thresholds. Finally, each alternative was evaluated against each criterion (and scenario, if applicable), checked for accuracy, and fed into the model to generate recommendations. In some cases, a sensitivity analysis on the criteria weightings was also conducted to examine the influence of different stakeholders on the recommendation. Model outputs were then discussed and compared to the counterfactuals (Fig. 1).

4.3. Limitations

Several limitations arose from this research study because of time, financial constraints, and availability of data. The selection of two organisations and application of MCDA using PROMETHEE does not allow generalisation of our findings to different organisational contexts and to other integrated assessment tools, and it seems likely that use cases emerge as a function of existing narratives, problem frames, knowledge of available tools, as well as data availability. However, this work may be a useful starting point for an expanded study involving a larger sample of organisations, country and sector contexts, alongside applications of a range of integrated decision support tools over a longer period of time, and a more systematic approach to selecting participating organisations. Sensitivity analysis of criteria weightings and preferences was not conducted in all use cases, model criteria weightings were assumed via stakeholder proxies in some instances, and ordinal data required expert input with a degree of subjectivity. The temporal nature of some of the use cases required multi-scenario analysis to be conducted in PROMETHEE as a workaround to a truly dynamic model.

5. Results and discussion

The following section details current shortcomings and challenges in ESG practices across both participating organisations (5.1), and introduces the PROMETHEE use cases, explaining the organisational context that lead to model development and any immediate effects the models had in terms of new insights, organisational understanding and decisions (5.2). Section 5.3 summarises qualitative feedback on the effectiveness of MCDA. In 5.4, we draw on observations, interviews, model development processes and diary entries to evaluate MCDA's impact in terms of balancing SDGs 8 and 13 in comparison to counterfactual decisions generated by business-as-usual decision support methods.

5.1. Current ESG practices, challenges & opportunities

Interviews with key stakeholders and observations of project and committee meetings revealed that both organisations were well-advanced in their corporate sustainability practices, but also experienced challenges. Both had dedicated Sustainability Managers and respective teams focused on environmental and social outcomes, progress against selected SDGs, accountability and transparency to shareholders and customers, and prioritisation and delivery of sustainability roadmaps. Trade-offs in pursuing the goals were openly discussed and debated. Operational teams were aware of the importance of holistic outcomes, and there were established key performance indicators (KPIs) to ensure alignment of incentives to these outcomes. Both organisations had governance committees at the executive and board level equipped with relevant and current information on progress against clearly defined sustainability targets. Sustainability roadmaps for both companies were ambitious, baselined against the internationally recognised Science-Based Targets Initiative (SBTi), with commitments and progress published via annual integrated reports that adhere to rigorous standards such as the Global Reporting Initiative (GRI) and Dow Jones Sustainability Indices (DJSI). Unbiased, externally facilitated materiality assessments were conducted and published annually across a wide spectrum of stakeholders, to ensure holistic influences on company strategy beyond shareholder and investor expectations. Climate adaptation and resilience were actively planned over long-term horizons and were seen as key risks in both businesses. Finally, external expertise was routinely accessed to ensure quality inputs, and there was a high degree of maturity in their sustainability assessments, including a critical view of where they could do better.

Despite established sustainability practices, stakeholders in both organisations identified several challenges, and these were validated and augmented by researcher-led observations of meetings and workshops. Interviews and focus groups conducted with executive leaders, sustainability leads, and strategy and finance teams confirmed an acknowledgment of mostly intuitive decision-making beyond consideration of explicit economic metrics such as Net Present Value (NPV) and Internal Rate of Return (IRR), and use of simplistic cost-benefit or risk-opportunity analysis methods to evaluate decisions with implications for sustainability, resulting in a default emphasis on commercial outcomes, and the notion of sustainability efforts as a "cost of doing business" rather than integrated within existing "business-as-usual" activity. One of the organisations was in the process of augmenting their standard business case template in order to increase awareness of environmental and social trade-offs for new projects, and encourage discovery and debate before material capital investment decisions were approved. However, neither organisation had sophisticated tools or systems for decision support in ESG beyond Excel spreadsheets. There was awareness and discussion of trade-offs, but these were not quantified or evaluated alongside commercial return. Sustainability trade-offs were addressed only after the capital investment decision was made on financial merit.

Focus group discussions revealed that teams are siloed within functionally aligned organisational structures (e.g., IT, Finance, Legal etc.) and that systems-thinking for sustainability was not embedded as a way of working. Participants observed that this often resulted in fragmented decision making and inconsistency in the balance of importance given to non-economic criteria. Lack of adequate expertise and siloed functions necessitates reliance on external consultants which leads to increased costs, potentially unnecessarily complicated or biased inputs, and need for internal translation to contextualise the findings.

Interviews with executives of both organisations surfaced challenges in the processes involving carbon accounting, monitoring, tracking of progress, and coping with continuously evolving ESG reporting standards and requirements, resulting in frequent data requirement changes. There were externally verifiable and audited processes to manage emissions accounting and other sustainability targets, but teams

remarked that this came with significant cost and time. Both organisations' executives pointed out that sustainability targets were not well-represented in performance incentives for remuneration and other company policies, suggesting this would need to change with emerging investor ESG scrutiny standards.

From a technical perspective, sustainability managers at both organisations expressed three major factors that hamper decision support processes for ESG. First, disparity in time horizons for optimization of outcomes. Financial outcomes were measured over 6–12 months while time horizons for sustainability endeavours varied from 5 to 30 years, leading to uncertainty in valuing impact across these time horizons. Second, the teams' ability to influence emissions reductions were often limited when the bulk of those emissions were not in the organisations' direct control. For example, this was observed in the form of slower than desired adoption of sustainable practices by contractors or other suppliers up and down the value chain, or in the form of policy uncertainty and market intervention by government.

Possibly most critically, practitioners and stakeholders at both organisations emphasised the need for simplicity and usability of any decision support tools that are introduced into existing business processes. Interviewees expressed a need for better links between enterprise value and sustainability initiatives, easy-to-use user interfaces, and transparent ways of tracking progress. Furthermore, communicating actual progress against the SDGs by demonstrating impact continues to be an area for improvement, with lack of sophisticated methods to account for uncertainty of costs, risks, benefits. The lack of standardised sustainability targets for the private sector created ongoing monitoring and reporting challenges, pointing to a need for recognised methods to scale down national contributions and planetary boundaries to individual private sector entity level targets.

5.2. The development and effect of integrated assessment use cases

Three use cases were iteratively identified, developed and evaluated over the course of 2023 (Table 2), each discussed in turn below. Across the use cases, setting preference thresholds (defining minimum differences needed for one option to be clearly preferred over other options) and indifference thresholds (the maximum difference where your preference for two options are indifferent) was completed using the software guiding feature because this was too complex for participants to undertake independently. Iterating and verifying model inputs several times for quality and consistency helped to lend credence to the process and the robustness of the models. Setting criteria weights was relatively easily accomplished in discussions with key stakeholders. All use cases presented some deviation from the counterfactual preferences and adjusting weights helped participants understand how criteria weights impacted rankings. The varying temporal nature of ESG decisions posed challenges in use cases 1 and 3 but these were resolved by applying scenario-analyses to the time horizons, noting need for more robust ways to incorporate variation and uncertainty in model inputs. Two use cases (1 and 3) were portfolio problems, in that organisations sought to select, prioritise and sequence a group of actions. Synergies were addressed either through separation of the problematic into multiple scenarios (using timescales to separate the dependencies) or more simply, by removing pre-requisite actions out of the MCDA altogether (see Table 4 and 9).

Table 2
Overview of Use cases.

	Use Case 1: Carbon Offset Choices	Use Case 2: Materiality Issues Prioritisation for GRI Reporting	Use Case 3: Enterprise Strategy Prioritisation
Problematic	Rank and recommend the top 3 offset alternatives in each timeframe	Rank material issues relative to impact against GRI-specified criteria	Rank investment in key strategic capabilities against company's strategic capitals for each timeframe
Options	Offset Types: Nature-Based Avoidance Units, On/Off-shore Afforestation (AFF) Units | native vs. exotic, Renewable Energy (RE) Units, Blue Carbon, Soil Carbon, Direct Air Carbon Capture & Storage (DACCS)	Material Issues, identified via interviews with a cross-section of internal and external stakeholders	17 Strategic Capabilities/ Initiatives mapped to 5 Key Strategic Capitals
Scenarios	(1) 2024–26; (2) 2027–2030; (3) 2030–2033	(1) Negative Material Issues; (2) Positive Material Issues	(1) Horizon 1 2024–26; (2) Horizon 2 2027–2030; (3) Horizon 3 2030+

Case 1. Carbon Offset Choices over 10-year horizon.

The first use case chosen within Organisation 1 investigated the purchase of a portfolio of carbon offsets in the 2023/24 timeframe in order to counter their residual (unabated) emissions and achieve a net-zero position, allowing exploration of the applicability of MCDA at a project level, accounting for varying uncertainty over time horizons. The time horizon for this decision extended over a decade, and while the company's volume of offsets needed was reasonably well-known, there was uncertainty of costs and availability of sequestration capacity over this time horizon, and clear trade-offs between economic (cost) and environmental (% of GHG demand met in the timeframe) criteria (Table 3). Furthermore, there were co-benefits (such as air/water quality, biodiversity, community impact etc.) as well as risks (credibility, accounting transparency, brand reputation etc.) that needed to be considered for each alternative (Table 3). To overcome the time horizon aspect, the model was set up over three scenarios, where each scenario represented a 3-year time window.

Each offset type was carefully evaluated on each criterion based on industry reference material, broker quotes, and internal subject matter expertise. Assumptions and uncertainties in estimates were also noted. For example, for offset types such as Blue Carbon that are still in nascent

Table 3
Criteria for assessing carbon offset choices over a 10 year period (Use Case 1).

Group (Weight)	Criterion	Unit
Economic (20%)	• Cost per Credit	NZD/offset
Environmental (20%)	• % of GHG demand met in the timeframe	1-5 Scale, mapped to tranches from 0 to 100%
Environmental Co-benefits (15%)	• Biodiversity • Water Consumption/ Quality • Air Quality • Climate Resilience • Natural Resource Intensity	1-5 Scale for each
Social Co-benefits (15%)	• Partnerships with Indigenous Groups/Iwi • Community Impact	1-5 Scale for each
Credibility (20%)	• Alignment to latest standards & guidelines • Certainty on claims • Monitoring, reporting, feasibility • Public Perception of the Offset • Brand/Reputation	1-5 Scale for each
Risks (10%)	• Tech readiness level • Political/Regulatory exposure • Certainty of cost (price volatility) • FTE effort & complexity to execute	High | Medium | Low

technology, we used an ordinal scale denoting technological readiness relative to more established technologies instead of absolute values. Sensitivity tests were run on criteria weights as well as costs in order to account for uncertainty in ETS carbon prices. Evaluations of the alternatives across a standard set of criteria varied within each of the scenarios, resulting in three ranking recommendations, which then formed the basis for a portfolio of offsets over the decade (Table 4).

The MCDA model produced a credible recommendation of the top 5 offset choices in each time horizon, which contributed in two ways; it helped to clarify the counterfactual position, and enabled separate evaluation of short and long-term horizons, resulting in a better informed "offset strategy" for the organisation (Table 4). The counterfactual decision for offsets procurement was company-owned forestry assets, despite this form of native afforestation having higher cost per credit compared to several other alternatives and not being immediately available to offset carbon emissions. The MCDA model showed that a high indifference threshold for cost per credit would be necessary to justify the organisational preference for company-owned forestry assets. Discussions revealed this was driven by familiarity with the existing portfolio as well as perceptions that NZ-based company owned land and forest were more credible, with greater accountability, and minimal reputational risk compared to credits purchased via the Emission Trading Scheme or offshore credits. A sensitivity analysis of the MCDA model showed that it would require a ~90% weighting of the credibility criterion to mimic the counterfactual ranking. However, since company-owned forest-based credits take considerable time to plant and mature to the point of being able to sequester adequate CO_2, the model pointed to a need for more immediate options to meet short-term demand. The consensus strategy was therefore to focus on a mix of offshore non-forestry and NZ forestry offsets in the short term in order to meet residual GHG emissions offset demand, followed by a longer-term intentional shift to company owned NZ-based forestry options.

Case 2. Prioritising Materiality Issues for GRI Reporting.

A second use case was surfaced by the Sustainability Team who were seeking better ways to plan and monitor action on 'material issues' - the organisation's most significant impacts on economy, environment and people. The team had used a third party to conduct annual materiality interviews with a cross-section of stakeholders within and external to the company in order to report material issues to investors in accordance with GRI standards, but now sought explicit ways to prioritise material issues, evaluate them against SDG targets, and track impact of actions to address them. The counterfactual approach they were using was a spreadsheet model rating each alternative on a scale of 1–10 for each of the 4 GRI criteria; scale, scope and irremediality were weighted equally, and multiplied by the likelihood criterion value to get a final rank score. An MCDA model was developed to rank the issues against GRI criteria for impact, with careful consideration of SDG indicators and thresholds to demonstrate and report progress to the public (Table 5), differing from the counterfactual in how the criteria were offset against each other. Material issues were divided into positive and negative impacts for the purposes of ranking and grouped into Environmental, Commercial, People, and Technology categories (Tables 6 and 7). The SDGs were used to develop indicators specific to the organisation in order to prioritise remediation actions for each material issue as well as report on annual progress.

The MCDA model rankings were very similar to the counterfactual rankings, because the underlying criteria and inputs were consistent across both methods. The key difference in rank score computation was the application of a weighted (in MCDA) vs. multiplier (in counterfactual) effect for the likelihood of an issue occurring, which meant that issues with low likelihood such as Risks from Infrastructure Safety were placed higher in rank under MCDA than under the counterfactual.

Case 3. Enterprise Strategy Prioritisation.

Table 4
Model ranking for use Case 1, AFF = afforestation; ETS = emissions trading Scheme; NZU = NZ units.

Option Rank	Scenario 2023-5	Scenario 2026-28	Scenario 2029+
1	Clean Cookstoves	Clean Cookstoves	AFF NZU Native
2	AFF Offshore	Owned/Mixed Native/Exotic	AFF NZU Mixed Native/Exotic
3	Renewable Energy Offshore	AFF NZU Mixed Native/Exotic	Clean Cookstoves
4	AFF NZU Native	AFF NZU Native	Owned NZ ETS Native Only
5	AFF NZU Mixed Native/Exotic	AFF Offshore	Owned Mixed Native/Exotic

Table 5
GRI Criteria used to rank material issues according to impact (Use Case 2).

Group (Weight)	Criterion	Unit
Scale (25%)	• Scale (Gravity/Consequence of Impact)	Scale 1-10
Scope (25%)	• Scope (Breadth of Impact)	Scale 1-10
Irremediality (25%)	• Irremediality (relevant for negative issues only)	Scale 1-10
Likelihood (25%)	• Likelihood (a higher likelihood weights the issue higher)	Percentage (Actual = 100%)

Table 6
Use Case 2 - Negative material issues ranking.

Counterfactual Ranking		Model Ranking	
Environment	Reduced river flows and water quality from diversion	Environment	Reduced river flows and water quality from diversion
Environment	Harm to freshwater biodiversity	Environment	Harm to freshwater biodiversity
Commercial	Erosion of public and customer trust due to market behaviour and pricing	Technology	Risks arising from infrastructure safety
Environment	Physical risks of a changing climate	Commercial	Erosion of public and customer trust due to market behaviour and pricing
Social	Adverse effects of generation assets and activities on cultural values	Environment	Physical risks of a changing climate
Social	Impacts on energy hardship	Commercial	Impacts of supply chain/ethical sourcing
Commercial	Impacts of supply chain/ethical sourcing	Social	Adverse effects of generation assets and activities on cultural values
Environment	Waste disposal and emissions	Social	Impacts on energy hardship
Commercial	Business performance: Diversity and equal opportunity	Technology	Business disruption, cyber-security, privacy breaches
Technology	Business disruption, cyber-security, privacy breaches	Environment	Waste disposal and emissions
Social	Negative impact on wellbeing of employees and contractors	Commercial	Business performance: Diversity and equal opportunity
Social	Interference with community recreational opportunities	Commercial	Good governance, ethical behaviour, and reporting
Commercial	Good governance, ethical behaviour, and reporting	Social	Negative impact on wellbeing of employees and contractors
Technology	Risks arising from infrastructure safety	Social	Interference with community recreational opportunities

Table 7
Use Case 2 - Positive material issues ranking.

Counterfactual Ranking		Model Ranking	
Environment	100% Renewable energy generation	Environment	100% Renewable energy generation
Commercial	Increasing supply of RE and storage	Commercial	Increasing supply of RE and storage
Environment	Reducing emissions of others (creating products)	Environment	Reducing emissions of others (creating products)
Environment	Improving biodiversity on land	Social	Leading and influencing change and progress on sustainability issues
Social	Leading and influencing change and progress on sustainability issues	Environment	Improving biodiversity on land
Social	Supporting opportunities for local communities	Social	Supporting opportunities for local communities
Social	Addressing inequality of access to new energy technology	Technology	Maximising benefits of demand flexibility and virtual power plants to support increased electrification
Technology	Maximising benefits of demand flexibility and virtual power plants to support increased electrification	Commercial	Policy changes that enable the rapid transition to a low carbon energy future
Commercial	Policy changes that enable the rapid transition to a low carbon energy future	Social	Addressing inequality of access to new energy technology

Organisation 2 had identified 17 distinct strategic initiatives which had been ranked by leadership according to perceived priority and urgency for the business. They were now looking to validate these investment choices using a more in-depth assessment of the impact of the strategic initiatives on a set of capitals across six areas of the business (commercial, customers, people, partnerships, environment, and networks). This allowed exploration of the applicability of MCDA at an enterprise strategy level, accounting for varying criteria weighting over time horizons. A potential application of MCDA was iteratively developed starting by articulating the problem statement, ascertaining the right level of granularity for options ranking, and weaving in the capitals as criteria in addition to risks and co-benefits in order to bring more credibility and transparency to the process.

Each strategic initiative was discussed for evaluation with respective business owners and calibrated with the help of the Finance team to ensure fair relativity across alternatives. Interval categories were used to guide participants in their assessment of alternative strategic initiatives against each criterion. The alternatives for this organisation's strategy were enumerated in the form of strategic 'capabilities' rather than distinct 'business initiatives', and hence estimating their commercial value in absolute terms was not possible. It was easier for participants to instead rank them in a relative manner by subjectively comparing the long term derivable value of each one. Arguably this resulted in loss of fidelity from the model's standpoint but was the most practical way to ensure a commercial lens still remained. This led participants to question the basis behind the counterfactual decisions as well. Criteria weights for Horizons 1, 2 and 3 were discussed relative to organisational targets and proposed by stakeholders for transparency, with a sensitivity analysis informing the final Commercial criterion weight (Table 8). Outliers were re-examined for accuracy. Three initiatives were removed from the Horizon 1 ranking as prerequisites but added back in for subsequent phases.

The lack of any variation in the inputs across alternatives caused one criterion to be dropped from the model – which itself generated good conversation between the participants. Compared to counterfactual 'intuitive' ranking, MCDA ranked initiatives performing well on economic criteria (such as Continuous cost and value management) lower than other initiatives that scored better on other criteria, revealing an implicit bias towards economic criteria in intuitive ranking (Table 9). A few alternatives were ranked differently by the model as a result of incorporating co-benefits and risks, suggesting that the model presented a more holistic enterprise view than the counterfactual (Table 9).

Table 8
Criteria for enterprise strategy prioritisation (Use Case 3).

Group (Weight H1; H2; H3)	Criterion	Unit
Commercial (45% | 36% | 27%)	• Annual earnings impact potential	1-5 Scale, mapped to tranches
Customer (18% | 24% | 24%)	• Effect on Net Promoter Score	1-5 Scale (1 = worst, 3 = neutral, 5 = best)
People (9% | 9% | 9%)	• Effect on Employee Engagement	1-5 Scale (1 = worst, 3 = neutral, 5 = best)
Environment (4% | 9% | 18%)	• Impact on GHG emissions	1-5 Scale (1 = worst, 3 = neutral, 5 = best)
Networks (14% | 14% | 14%)	• Effect on Network Resilience	1-5 Scale (1 = worst, 3 = neutral, 5 = best)
Risks (5%)	• Capability/Readiness level • Political/Social License • Indirect Cost/ Complexity	High | Medium | Low
Co-benefits (5%)	• Data & Cyber Maturity • Health & Safety Maturity • Brand/Reputation • Business Unit Integration	High | Medium | Low

5.3. *Effectiveness of MCDA*

Over the course of this study, participating sustainability leads and team members progressively learned how to develop the problematic and determine model parameters and inputs, and it became easier over time and with experience to identify potentially suitable problems, communicate and develop models. Qualitative feedback on ease of use, relevance and future potential of MCDA was largely positive across both organisations. Key contributions identified were that "it forced [teams] to calibrate their initiatives transparently"; "it exposed previously held biases and made explicit the reasons for those biases", and "it helped to integrate our thinking across functional teams". Interview feedback confirmed that MCDA concepts were easily understood, although details of criteria weightings and indifference thresholds were harder to comprehend. Ensuring fidelity of criteria (no duplication or inadvertent suppression of problem dimensions) and satisfaction with weights took iteration. Sensitivity analyses on different weights greatly improved stakeholders' understanding and comfort with model outcomes. Once the models had been established, the organisation's subject matter experts were able to collate inputs without difficulty especially where the criteria overlapped with counterfactual business cases, such as economic impact and standard environmental measures such as GHG emissions. When outcomes were communicated and compared with the counterfactual, good questions were raised around the reasons for the differences, the sensitivities of criteria weights, and fidelity of inputs into the models. This indicated a high level of understanding of the process and acknowledgment of its value across the use cases. However, deploying and validating the models took time.

MCDA models rely on accurate inputs based on credible sources or forecasted impacts for each alternative, and one of the primary challenges in setting up the model values was establishment of units of measurement for each criterion because data availability for non-economic measures such as biodiversity, water quality, social co-

Table 9
Model ranking for Strategy prioritisation (Use Case 3).

	Horizon 1	Horizon 2	Horizon 3
Counterfactual	MCDA		
8. Enabling Data Insights	Prerequisites: 7.Mature Data, Digital & Technology Practices 8. Enabling Data Insights 11. Cyber Security Needs 17. Digitised and data-driven network		
12. Cost and value management	1. Design and develop new products and solutions	1. Design and develop new products and solutions	1. Design and develop new products and solutions
16. Fit for Purpose Infrastructure (Logistics and Commercial models)	16. Fit for Purpose Infrastructure (Logistics and Commercial models)	16. Fit for Purpose Infrastructure (Logistics and Commercial models)	16. Fit for Purpose Infrastructure (Logistics and Commercial models)
17. Digitised and data-driven network	4. Be the preferred partners and employer (brand and culture)	4. Be the preferred partners and employer (brand and culture)	4. Be the preferred partners and employer (brand and culture)
3. Customer Experience (self-service)	15. Automated Processing	14. Operational and Safety Excellence	14. Operational and Safety Excellence
14. Operational and Safety Excellence	10. Systems Integration	8. Enabling Data Insights	8. Enabling Data Insights
15. Automated Processing	2. Strategic relationship management	6. Low Emissions fleet	6. Low Emissions fleet
1. Design and develop new products and solutions	3. Customer Experience (self-service)	15. Automated Processing	15. Automated Processing
2. Strategic relationship management	12. Cost and value management	2. Strategic relationship management	2. Strategic relationship management
4. Be the preferred partners and employer (brand and culture)	14. Operational and Safety Excellence	10. Systems Integration	3. Customer Experience (self-service)
5. Circular solutions	6. Low Emissions fleet	3. Customer Experience (self-service)	5. Circular solutions
6. Low Emissions fleet	5. Circular solutions	12. Cost and value management	10. Systems Integration
7. Mature data, Digital & Technology Practice	13. Workforce alignment	11. Cyber Security Needs	12. Cost and value management
9. Streamlined business Processes/ Platforms	9. Streamlined business Processes/ Platforms	5. Circular solutions	7. Mature data, Digital & Technology Practice
10. Systems Integration		7. Mature data, Digital & Technology Practice	13. Workforce alignment
11. Cyber Security Needs		13. Workforce alignment	9. Streamlined business Processes/ Platforms
13. Workforce alignment		9. Streamlined business Processes/ Platforms	11. Cyber Security Needs

benefits, infrastructure readiness, was poor. Furthermore, it became evident during model development that there was a trade-off between complexity in pursuing estimations in absolute measures and fidelity of inputs. In many cases we resorted to ordinal or interval scales that allowed participants to assess each alternative against each criterion in a consistent way. The limitations of this approach were a) the loss of quality in the inputs; and b) the impact on indifference thresholds due to step changes in the scoring compared to more continuous absolute values. This pointed to a need to baseline and monitor absolute values where measurable if MCDA is to be scaled more widely. This study did not extend to monitoring and management of the models, however, both organisations acknowledged the necessity of ensuring the input data are kept up to date as market conditions, policy and technology uncertainty evolve, and the consequent need for specialised resources, information management technology and infrastructure to measure, monitor and store this information and enable good data governance. Across use cases, project level decisions were more dependent on numerical data for their ranking outcomes than strategy level decisions, which are likely to rely largely on ordinal or interval-based datasets.

Taken together, we find that MCDA was useable, intuitive, widely understood, that models were easily developed by collaborating with relevant staff to identify alternative options for ranking, appropriate criteria, measures, and parameters across all three use cases. MCDA was effectively applied to decision problems ranging from ranking of materiality of sustainability issues, investment decisions in sustainable alternatives and enterprise level capability prioritisation. Use cases contributed by aiding the development of the sustainability strategy to account for varying degrees of certainty in short, mid, and long-term time horizons, influencing the narrative on preferred (counterfactual) decisions as a result of explicit criteria weightings exposing (and in some cases, correcting) implicit biases and by involving a wider group of stakeholder participants and deliberation of a wider range of inputs than the status quo, bringing clarity to goals, assumptions and preferences. In one of the organisations some team members found MCDA so promising that they started bringing forward potential use cases for experimentation and trial of their own accord.

Strategy level use cases were more ambiguous and complex to model than project level use cases both in terms of problematic as well as generation of input data. Enterprise level decision support spanned more stakeholders, so aligning criteria and weights was more challenging, as was collating and calibrating inputs across the alternatives. As strategic decisions typically rely on more intuition and experience of senior management, the outcomes produced by the model for Use Case 3 were less aligned to outcomes of counterfactual approaches, and more difficult for participants to accept. For example, one of the alternatives in Use Case 3 ranked lower than in the counterfactual, because economic criteria had effectively been weighted at over 90% (implicitly) in the intuitive investment allocation. While broader stakeholder participation was more challenging to manage, the breadth of stakeholder participation was also consistently cited as a marked improvement of the MCDA process compared to the status quo, due to the need to explicitly identify criteria, measures, and preferences in the MCDA model construction process. Participants could see value in how analyses of this kind benefitted from diverse viewpoints, as the models were able to demonstrate how decisions were balanced across multiple dimensions that different stakeholders considered more or less important than others. Despite these challenges, stakeholders found the enterprise-level trials useful for better consistency, accountability, and impact reporting, and all stakeholders interviewed stated they would consider incorporating a more robust and transparent approach such as MCDA to augment or assist in forming the basis for executive decisions.

5.4. Contribution towards balancing SDGs 8 and 13

The MCDA use cases in this study weighted environmental and social criteria (and co-benefits), included a larger group of stakeholders in

assembling inputs, and ranked enterprise strategies according to impact, integrating business innovation, technology, shareholder returns, community values, regulatory policy, and infrastructure dimensions at an operational and practical level. Use cases 1 and 3 explicitly analysed the alternative options using carefully considered weights and indifference thresholds between environmental and economic criteria. Taken together, MCDA results across all cases favoured alternatives that were more 'balanced' towards environmental outcomes as a result of including additional non-economic criteria in the evaluation. Use case 1 (Carbon Offset choices) recommended environmentally achievable outcomes over a multi-year horizon, and showed that the counterfactual would not have been able to offset GHG emissions in the short term, but instead overweighted credibility over cost. Furthermore, Use case 3 resulted in prioritisation of strategic initiatives with a high emissions reductions impact.

Use case 2 aimed at ranking material sustainability issues according to impact. This use case demonstrated potential for more comprehensively evaluating enterprise strategy and capital investment allocation against environmental and climate risks, social commitments, and enhancing the companies' ability to provide verifiable and transparent disclosure of impact and actions taken to mitigate material issues. In use cases 1 and 3, developing the model necessitated a much wider assessment of inputs and a higher degree of integration in the assessments when compared to the counterfactual processes organisations had in place, which were primarily one-dimensional. Although forcing criteria weights across the trade-offs challenged the model development process, we can infer based on these use cases that MCDA has the potential to support the "intentional shifts" that are preconditions for transformative pathways (Marshall et al., 2022; Sandberg and Tienari, 2022; van Tulder and van Mil, 2023). While MCDA application alone cannot reorient business to achieve green or circular growth models, a more fulsome criteria set with unbiased weightings can reorient counterfactual economic growth-led approaches to take a more balanced view.

Qualitative feedback from both organisations validated the improvement in robustness of the process of decision-making and the explicit nature of the assumptions made in selecting criteria and corresponding weights, which increased transparency of the outcomes, improved the decision-making process by integrating disparate criteria into one single assessment framework, and also improved outcomes by incorporating environmental emissions impacts and challenging implicit cost minimisation (or profit maximisation) assumptions.

6. Conclusions

While it was not possible to trace the influence of use cases on sustainability performance in the study time frame, the use cases in this study demonstrate that the application of MCDA models has the potential to broaden the meaning of 'value' to encompass criteria beyond traditional economic measures, and facilitate a shift from win-wins or trade-offs to more holistic and integrated practices, contributing to the creation of new, transformative pathways for decision-making. MCDA contributed to addressing the "co-benefits failure" (Stern et al., 2022) by considering benefits beyond market rewards as consequence of inclusion of ecosystem health, biodiversity, health, etc. in policy as well as corporate decision-making (as seen in the example of use cases 1 and 3). Ultimately, the use of an outranking-based MCDA allowed for uncertain and multi-dimensional decision problems to be decomposed into ordinal rank recommendations. While this might seem like an over-simplification, the exercise proved valuable in exposing assumptions (e.g., inherent strategic biases in use cases 1 and 3), incorporating more robust ranking methods (use case 2), and providing a useful baseline from which to evaluate next steps, thus also enhancing legitimacy of the process and its results.

Although deploying MCDA at a corporate level is not straightforward, the challenges outlined here can be overcome with the right skills, organisational culture for change, enhanced data infrastructure and process articulation. The MCDA trials raised discussion about further future potential applications for MCDA in both organisations. Ideas raised included incorporating traditional CBA (i.e., NPV) into such analyses and augmenting existing investment case approval processes to evaluate business cases using MCDA model ranking recommendations. Applying the methodology more widely at the two organisations would require resourcing a subject matter expert in MCDA to lead inquiry, building and maintenance of the models, productising model templates for a range of different use cases to avoid reconstruction, and capturing assumptions made in scoring alternatives. Once constructed, the MCDA models have the advantage of being able to be run repeatedly as inputs or market conditions change, making the overall decision-making workload more efficient. MCDA should augment existing processes rather than run separately from them, to ensure consistency of participation and leverage of existing data or analysis. Finally, enterprise level software and information management technology will need to be identified or developed and processes defined and followed for good governance and application of methods.

This research illustrates the potential of MCDA in enhancing corporate ESG practices by providing a practical framework for integrating incommensurate sustainability criteria into complex decision-making. The findings suggest integrated assessment tools play an important role for reframing how companies can achieve shareholder returns and sustainable outcomes in an integrated manner. For any transformative gains to be realised via MCDA, companies must be willing to address and assess non-economic long-term risks, redefine shareholder value, and evaluate their investment options and actions across diverse measures, involving diverse stakeholder groups along the way.

Ethics statement

This project has been evaluated by peer review and judged to be low risk. Consequently, it has not been reviewed by one of the University's Human Ethics Committees. The authors are responsible for the ethical conduct of this research. The following actions have been taken to mitigate ethical breaches. The scope of work as a researcher vis-à-vis other duties was discussed and approved by the Chairs and Chief Executives of both organisations participating in this study. All forums and media of data collection included explicit and upfront declaration as to the purpose and use of the information. Any commercial sensitive or confidential information was anonymised in the final research report and presentation. No participants in the study were individually identified, and any qualitative points of view aggregated before presentation. No individually identifiable data was captured as part of this research.

CRediT authorship contribution statement

N. Sanatkumar: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **A.L. Berka-Harnmeijer:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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